

Visualizing the Quantum Ising Phase Transition

From Ferromagnetic Order to a Quantum Paramagnet

Physical background and starting point

Consider N spin-1/2 sites on a ring, with Hamiltonian

$$H(g) = -J \sum_{j=1}^N \sigma_j^z \sigma_{j+1}^z - Jg \sum_{j=1}^N \sigma_j^x, \quad \sigma_{N+1}^\alpha = \sigma_1^\alpha, \quad g = \frac{h}{J}. \quad (1)$$

Set $J = 1$ and work at zero temperature. Here $h \geq 0$ is the transverse field and the operators are Pauli matrices, with eigenvalues ± 1 , rather than spin operators $\sigma^\alpha/2$.

The interaction favours spins aligned along z . At weak transverse field, spins at distant sites remain strongly correlated in this direction: the system has ferromagnetic order in the infinite-chain limit. At strong field, the ground state approaches

$$|+x\rangle^{\otimes N}, \quad |+x\rangle = \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}}. \quad (2)$$

This state is a *quantum paramagnet*: it has no spontaneous z order, but is polarized along the applied x field. It is not a thermal mixture of randomly oriented spins.

For this normalization the infinite chain has its zero-temperature quantum critical point at $g_c = 1$. A finite chain instead exhibits a rounded change. Your task is to visualize both regimes and show how the crossover sharpens with increasing N , rather than manufacture a discontinuity in a small system.

Suggested parameters: use $N = 6, 8, 10$ and, 18. Sweep g from 0.1 to 3, with extra points near $0.7 \leq g \leq 1.3$. Use a spacing around 0.02 near the transition as an initial choice, then check sensitivity to that spacing. Precompute the data before animating it.

Construct the ground states and order diagnostics

A. Build the Hamiltonian

Encode the 2^N spin configurations by binary integers b . Let bit $b_j = 0$ represent spin up and $b_j = 1$ spin down at site j , and define $z_j(b) = 1 - 2b_j$. Then

$$H_{bb} = -J \sum_j z_j(b) z_{j+1}(b), \quad H_{b',b} = -Jg \quad \text{if } b' \text{ differs from } b \text{ by one bit flip}. \quad (3)$$

Count each ring bond once. If you generate every bit-flip connection from every basis state, both matrix directions are already present; do not add the transpose a second time. Use a sparse matrix and solve for the lowest *algebraic* eigenvalue, not the eigenvalue nearest zero:

$$H(g) |\psi_0(g)\rangle = E_0(g) |\psi_0(g)\rangle. \quad (4)$$

B. Handle the finite-system symmetry correctly

The Hamiltonian commutes with $P = \prod_j \sigma_j^x$, which reverses all z spins. For $g > 0$, the finite-ring ground state has even parity, so $\langle \sigma_j^z \rangle = 0$ even in the regime corresponding to a ferromagnet. **Zero mean longitudinal magnetization therefore does not identify the paramagnetic phase.**

At small g , the lowest even and odd states become nearly degenerate. A reliable approach is to diagonalize in the even basis

$$|b; +\rangle = \frac{|b\rangle + |\bar{b}\rangle}{\sqrt{2}}, \quad b < \bar{b}, \quad (5)$$

where $\bar{b}$ flips every bit. If Q contains these basis vectors as columns, use $H_+ = Q^\dagger H Q$ and reconstruct $\psi_0 = Q\psi_{0,+}$. Verify $\langle P \rangle \simeq 1$; a symmetry-breaking numerical mixture must not be mistaken for a selected ferromagnetic vacuum.

C. Compute quantities that distinguish the regimes

For each field and size calculate

$$M_z = \frac{1}{N} \sum_j \sigma_j^z, \quad m_{\text{rms}}(g) = \sqrt{\langle M_z^2 \rangle}, \quad (6)$$

$$m_x(g) = \frac{1}{N} \sum_j \langle \sigma_j^x \rangle, \quad G_{ij}(g) = \langle \sigma_i^z \sigma_j^z \rangle. \quad (7)$$

All expectations are in $\psi_0(g)$. The root-mean-square longitudinal magnetization detects ordering without selecting one sign. At strong field, $m_{\text{rms}} \rightarrow 1/\sqrt{N}$, not exactly zero for finite N . The diagonal entries $G_{ii} = 1$ are trivial; ordering is diagnosed by *off-diagonal*, especially long-distance, correlations.

Also compute the exact probability of measuring a total magnetization m :

$$p_g(m) = \sum_{b: N^{-1} \sum_j z_j(b)=m} |\psi_{0,b}(g)|^2, \quad m = -1, -1 + \frac{2}{N}, \dots, 1. \quad (8)$$

The z observables can be evaluated directly from basis probabilities. For m_x , sum products of amplitudes connected by one-bit flips.

Make the transition visual (Most of these instructions are for AI)

Create a **single display** containing the quantum results below and the temperature-controlled animation specified in the next subsection. The four quantum panels share a field slider g and use the computed ground state at that field. Keep their field control distinct from the animation temperature.

1. **Spin measurement strips.** Show several rows of N coloured cells, with two colours representing the outcomes $\uparrow$ and $\downarrow$ of a simultaneous z measurement. Draw configurations directly from the exact probabilities $|\psi_{0,b}|^2$. These are independent samples, not a Monte Carlo chain. Label them “possible measurement outcomes”, not “the quantum state”. At weak field, most rows should be strongly aligned; at strong field, mixed outcomes become common even though the state is polarized along x .
2. **Correlation heatmap.** Display G_{ij} with fixed colour limits $[-1, 1]$, a colour bar, and site labels. Show how strong correlations across the ring are suppressed as the field increases. Do not rescale the colour range separately at every field.

3. **Magnetization probability distribution.** Display $p_g(m)$ as a bar chart on fixed axes $-1 \leq m \leq 1$. Show the change from two peaks near opposite ordered magnetizations to a distribution concentrated nearer zero. Use exact probabilities here, not a noisy histogram of a few sampled strips.
4. **Order and polarization curves.** Plot $m_{\text{rms}}(g)$ and $m_x(g)$, with a moving marker at the current field. Include curves for the chosen system sizes and label g , N , and the two observables clearly.

The slider must allow movement from weak to strong field and back. Since these are equilibrium states, reversing the slider should reproduce the same observables, apart from newly drawn measurement samples. Random measurement outcomes are illustrations of a quantum probability distribution; they are not real-time spin-flip trajectories.

Save three representative frames near $g = 0.2$, $g = 1$, and $g = 3$. Include the numerical values of m_{rms} , m_x , and the mean diametrically opposite correlation

$$G_{\text{far}}(g) = \frac{1}{N} \sum_j G_{j, j+N/2}(g). \quad (9)$$

Use the same visual scales so that the frames can be compared directly.

Temperature-controlled spin animation (visual component)

Include a simple animated spin lattice showing a disordered paramagnetic state becoming ordered on cooling and disordered again on heating. This is a visual component of the same submission; no additional derivation of a simulation algorithm is required.

For this panel, use the **classical two-dimensional Ising model** on a 32×32 square lattice with periodic boundaries and no external field:

$$E = -J \sum_{\langle ij \rangle} s_i s_j, \quad s_i = \pm 1, \quad J = k_B = 1. \quad (10)$$

Label the panel “Thermal Ising animation”. It illustrates temperature-driven ordering and is distinct from the zero-temperature quantum chain above; the temperature T must not be identified with the transverse field g .

1. **Show the spins.** Represent up and down spins by two colours, with arrows if useful. Start from a random configuration and retain the current configuration as the animation runs.
2. **Add a temperature slider.** Let the user vary T between 0.5 and 5. Mark the infinite-square-lattice critical temperature $T_c = 2/\ln(1 + \sqrt{2}) \simeq 2.269$. At low temperature, allow aligned domains to form and grow; at high temperature, show fluctuating, disordered patterns. Do not force an instantaneous visual switch at T_c .
3. **Animate local thermal updates.** AI-generated code may use a standard heat-bath or Metropolis rule. For example, a heat-bath update chooses a random site and sets it to $+1$ with probability

$$P(s_i = +1 \mid \{s_j\}_{j \sim i}) = \frac{1}{1 + \exp[-2 \sum_{j \sim i} s_j / T]}, \quad (11)$$

and to -1 otherwise, where the sum runs over its four neighbours. Perform enough updates between frames to make domain growth visible. Keep updating at the selected temperature when the slider stops moving.

4. **Provide simple controls and readouts.** Include play/pause, reset to random spins, and animation speed. Display the current temperature and the instantaneous magnetization magnitude $|m| = |L^{-2} \sum_i s_i|$ with $L = 32$.

Required demonstration: start near $T = 4$, lower the temperature to about $T = 1.5$, wait for ordered domains to grow, then return to $T = 4$ and show their breakup. Preserve the spin configuration when changing T ; only the reset button should randomize it. Repeat if necessary: domain growth is stochastic and global alignment can take time, especially near criticality or when large competing domains remain.

Submit the animation as part of the notebook or interactive display, with one low-temperature and one high-temperature snapshot. The thermal updates provide a visual relaxation process; frame count is not a calibrated physical time. No quantum time-evolution calculation, detailed Monte Carlo analysis, or separate phase-transition measurement is required for this panel.

Locate the transition region from the data

Compute the transverse response

$$\chi_x(g) = \frac{dm_x}{dg} \simeq \frac{m_x(g + \Delta g) - m_x(g - \Delta g)}{2\Delta g}. \quad (12)$$

Use this expression on a locally uniform grid, or a differentiation formula that accounts for nonuniform spacing. For each N , locate the peak in the region around $g = 1$ and report its position $g_*(N)$. Refine the field grid near that peak and report the change in its estimated position.

Compare $g_*(N)$ with 1 and examine how the peak and order curves change with size. The peak is a finite-size indicator, not an exact critical point. Do not fit an unsupported extrapolation from a few small chains or identify g_* with the field where m_{rms} vanishes: it never vanishes exactly in this finite system.

Required physical and numerical checks

Weak-field and strong-field limits

At $g = 0$, use the even ground state explicitly for the benchmark:

$$|\psi_F\rangle = \frac{|\uparrow \cdots \uparrow\rangle + |\downarrow \cdots \downarrow\rangle}{\sqrt{2}}. \quad (13)$$

Verify $E_0 = -JN$, $m_{\text{rms}} = 1$, $m_x = 0$, $G_{ij} = 1$, and $p(1) = p(-1) = 1/2$. Explain why its mean z magnetization is nevertheless zero.

For an exact independent-spin test, set $J = 0$ while holding $h > 0$ in the original form $H = -J \sum_j \sigma_j^z \sigma_{j+1}^z - h \sum_j \sigma_j^x$. The ratio $g = h/J$ is not used in this separate test. Verify

$$E_0 = -hN, \quad m_x = 1, \quad m_{\text{rms}} = N^{-1/2}, \quad G_{ij} = \delta_{ij}. \quad (14)$$

For k measured down spins,

$$p\left(m = 1 - \frac{2k}{N}\right) = 2^{-N} \binom{N}{k}. \quad (15)$$

At finite $g = 3$, the interacting result only approximates this limit; do not impose exact agreement with it.

Numerical reliability

- Check $H = H^\dagger$, $\langle \psi_0 | \psi_0 \rangle = 1$, and $\langle P \rangle = 1$. Report the eigenpair residual $\|H\psi_0 - E_0\psi_0\|/\max(1, |E_0|)$. A small residual alone does not prove that the selected eigenstate is the ground state.
- For a small system, compare the sparse result with a full diagonalization. Tighten solver tolerances and report changes in the observables.
- Check $\sum_m p_g(m) = 1$ and $\sum_m m^2 p_g(m) = m_{\text{rms}}^2$. For the even state, check $p_g(m) = p_g(-m)$.
- At several fields, verify the independent derivative relation

$$-\frac{1}{NJ} \frac{dE_0}{dg} = m_x(g). \quad (16)$$

Use a finite difference of the computed energies and refine its spacing.